

JAVA PATTERN PROGRAMS

Complete Reference Guide

With Code & Executed Outputs

19 Complete Pattern Solutions

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Star Patterns

Solid Square

Description: A simple solid square pattern made with asterisks

Code:

```
class Main {  
    public static void main(String[] args) {  
        int n = 4;  
        for (int i = 1; i<=n; i++){  
            for (int j = 1; j<=n; j++){  
                System.out.print("* ");  
            }  
            System.out.println();  
        }  
    }  
}
```

Output:

```
* * * *  
* * * *  
* * * *  
* * * *
```

Solid Rectangle

Description: A solid rectangle with custom rows and columns

Code:

```
class Main {  
    public static void main(String[] args) {  
        int n = 3;  
        for (int i = 1; i<=n; i++){  
            for (int j = 1; j<=5; j++){  
                System.out.print("* ");  
            }  
            System.out.println();  
        }  
    }  
}
```

Output:

```
* * * * *  
* * * * *  
* * * * *
```

Right Angle Triangle

Description: A right-angled triangle with asterisks, increasing in each row

Code:

```
class Main {
    public static void main(String[] args) {
        int n = 4;
        for (int i = 1; i<=n; i++){
            for (int j = 1; j<=i; j++){
                System.out.print("* ");
            }
            System.out.println();
        }
    }
}
```

Output:

```
*
* *
* * *
* * * *
```

Solid Rhombus

Description: A diamond-shaped rhombus pattern with spaces and asterisks

Code:

```
class Main {
    public static void main(String[] args) {
        int n = 5;
        for (int i = 1; i<=n; i++){
            for (int j = 1; j<=n-i; j++){
                System.out.print(" ");
            }
            for (int j = 1; j<=n; j++){
                System.out.print("* ");
            }
            System.out.println();
        }
    }
}
```

Output:

```
  * * * * *
 * * * * *
* * * * *
 * * * * *
  * * * * *
```

Inverted Right Angle Triangle

Description: A right-angled triangle that decreases in each row

Code:

```
class Main {
    public static void main(String[] args) {
        int n = 5;
        for (int i = 1; i<=n; i++){
            for (int j = 1; j<=n-i+1; j++){
                System.out.print("* ");
            }
            System.out.println();
        }
    }
}
```

Output:

```
* * * * *
* * * *
* * *
* *
*
```

Solid Pyramid Pattern

Description: A centered pyramid with increasing asterisks

Code:

```
class Main {
    public static void main(String[] args) {
        int n = 5;
        for (int i = 1; i<=n; i++){
            for (int j = 1; j<=n-i; j++){
                System.out.print(" ");
            }
            for (int j = 1; j<=2*i-1; j++){
                System.out.print("* ");
            }
            System.out.println();
        }
    }
}
```

Output:

```
      *
     * * *
    * * * * *
   * * * * * * *
  * * * * * * * *
```

Inverted Solid Pyramid Pattern

Description: A centered pyramid that decreases from bottom to top

Code:

```
class Main {
    public static void main(String[] args) {
        int n = 5;
        for (int i = 1; i<=n; i++){
            for (int j = 1; j<=i-1; j++){
                System.out.print("_ ");
            }
            for (int j = 1; j<=((2*n)-(2*i)+1); j++){
                System.out.print("* ");
            }
            System.out.println();
        }
    }
}
```

Output:

```
* * * * *
_ * * * *
_ _ * * *
_ _ _ * *
_ _ _ _ *
```

Hollow Rectangle Pattern

Description: A rectangle with asterisks only on borders

Code:

```
class Main {
    public static void main(String[] args) {
        int row = 5;
        int col = 10;
        for (int i = 1; i<=row; i++){
            for (int j = 1; j<=col; j++){
                if(i==1 || i == row ){
                    System.out.print("* ");
                }else if(j == 1 || j == col){
                    System.out.print("* ");
                }else{
                    System.out.print("  ");
                }
            }
            System.out.println();
        }
    }
}
```

Output:

```

* * * * *
*               *
*               *
*               *
* * * * *

```

Hollow Right Angle Triangle

Description: A right-angled triangle with hollow interior

Code:

```

class Main {
    public static void main(String[] args) {
        int row = 8;
        for (int i = 1; i<=row; i++){
            for (int j = 1; j<=i; j++){
                if(i==1 || i == 2 || i == row ){
                    System.out.print("* ");
                }
                else if(j == 1 || j == i){
                    System.out.print("* ");
                }else{
                    System.out.print("_ ");
                }
            }
            System.out.println();
        }
    }
}

```

Output:

```

*
* *
*  *
* _ *
* _ _ *
* _ _ _ *
* _ _ _ _ *
* _ _ _ _ _ *
* * * * *

```

Hollow Pyramid

Description: A centered pyramid with hollow interior

Code:

```

class Main {
    public static void main(String[] args) {
        int rowNo = 7;
        for (int i = 1; i<=rowNo; i++){
            for (int j = 1; j<=rowNo-i; j++){

```

```

        System.out.print(" ");
    }
    if(i==1 || i == rowNo){
        for (int j = 1; j<=2*i-1; j++){
            System.out.print("* ");
        }
    }
    else{
        System.out.print("* ");
        for (int j = 1; j<=2*i-3; j++){
            System.out.print(" ");
        }
        System.out.print("* ");
    }
    System.out.println();
}
}
}

```

Output:

```

      *
     * *
    *   *
   *     *
  *       *
 *         *
*           *
* * * * *

```

Diamond Pattern

Description: A complete diamond shape formed by asterisks

Code:

```

class Main {
    public static void main(String[] args) {
        int rowNo = 6;
        for (int i = 1; i<=rowNo; i++){
            for (int j = 1; j<=rowNo-i; j++){
                System.out.print(" ");
            }
            for (int j = 1; j<=2*i-1; j++){
                System.out.print("* ");
            }
            System.out.println();
        }
        for (int i = 1; i<=rowNo-1; i++){
            for (int j = 1; j<=i; j++){
                System.out.print(" ");
            }
            for (int j = 1; j<=2*rowNo-2*i; j++){
                System.out.print("* ");
            }
            System.out.println();
        }
    }
}

```

```

    }
}

```

Output:

```

      *
    * * *
  * * * * *
* * * * * * *
* * * * * * * * *
* * * * * * * * * *
  * * * * * * *
    * * * * *
      * * *
        * *

```

Hollow Diamond

Description: A diamond shape with hollow interior and border asterisks

Code:

```

class Main {
    public static void main(String[] args) {
        int rowNo = 7;
        for (int i = 1; i<=rowNo; i++){
            for (int j = 1; j<=rowNo-i; j++){
                System.out.print(" ");
            }
            if(i == 1){
                for (int j = 1; j<=2*i-1; j++){
                    System.out.print("* ");
                }
            }else{
                System.out.print("* ");
                for (int j = 1; j<=2*i-3; j++){
                    System.out.print("_ ");
                }
                System.out.print("* ");
            }
            System.out.println();
        }
    }
}

```

Output:

```

      *
    * _ *
  * _ _ _ *
* _ _ _ _ _ *
* _ _ _ _ _ _ _ *

```



```

      *           *
    * - - - - - *
    - - - - -

```

Butterfly Pattern

Description: Two triangles facing each other forming a butterfly shape

Code:

```

class Main {
    public static void main(String[] args) {
        int num = 4;
        for (int row=1; row<= num; row++){
            for (int col = 1 ; col <= row; col++){
                System.out.print("* ");
            }
            for (int col = 1 ; col <= 2*num-2*row; col++){
                System.out.print(" ");
            }
            for (int col = 1 ; col <= row; col++){
                System.out.print("* ");
            }
            System.out.println();
        }
        for (int row=1; row<= num; row++){
            for (int col = 1 ; col <= num-row+1; col++){
                System.out.print("* ");
            }
            for (int col = 1 ; col <= 2*row-2; col++){
                System.out.print(" ");
            }
            for (int col = 1 ; col <= num-row+1; col++){
                System.out.print("* ");
            }
            System.out.println();
        }
    }
}

```

Output:

```

      *           *
    * *         * *
  * * *       * * *
* * * * * * * *
* * * * * * * *
* * *       * * *
* *         * *
      *           *

```

Numerical Patterns

Numerical Pattern - Increasing

Description: Each row prints increasing numbers from 1

Code:

```
class Main {  
    public static void main(String[] args) {  
        int num = 5;  
        for (int row=1; row<= num; row++){  
            for (int col = 1 ; col <= row; col++){  
                System.out.print(col + " ");  
            }  
            System.out.println();  
        }  
    }  
}
```

Output:

```
1  
1 2  
1 2 3  
1 2 3 4  
1 2 3 4 5
```

Numerical Pyramid Pattern

Description: Numbers increase then decrease in pyramid form

Code:

```
class Main {  
    public static void main(String[] args) {  
        int num = 4;  
        for (int row=1; row<= num; row++){  
            for (int col = 1 ; col <= num-row; col++){  
                System.out.print(" ");  
            }  
            for (int col = 1 ; col <= row; col++){  
                System.out.print(col + " ");  
            }  
            for (int col = 1 ; col <= row-1; col++){  
                System.out.print(row-col + " ");  
            }  
            System.out.println();  
        }  
    }  
}
```

Output:

```
1
```

```
    1 2 1
   1 2 3 2 1
  1 2 3 4 3 2 1
```

Row Number Pattern

Description: Each row prints the row number repeated

Code:

```
class Main {
    public static void main(String[] args) {
        int num = 4;
        for (int row=1; row<= num; row++){
            for (int col = 1 ; col <= num-row; col++){
                System.out.print(" ");
            }
            for (int col = 1 ; col <= 2*row -1; col++){
                System.out.print(row + " ");
            }
            System.out.println();
        }
    }
}
```

Output:

```
    1
   2 2 2
  3 3 3 3 3
 4 4 4 4 4 4 4
```

Character Patterns

Character Pattern - Increasing

Description: Each row prints increasing characters from A

Code:

```
class Main {
    public static void main(String[] args) {
        int num = 5;
        for (int row=1; row<= num; row++){
            char character = 65;
            for (int col = 1 ; col <= row; col++){
                System.out.print(character + " ");
                character++;
            }
            System.out.println();
        }
    }
}
```

Output:

```
A
A B
A B C
A B C D
A B C D E
```

Character Pattern - Decreasing

Description: Each row prints decreasing characters

Code:

```
class Main {
    public static void main(String[] args) {
        int num = 5;
        for (int row=1; row<= num; row++){
            char character = 69;
            for (int col = 1 ; col <= row; col++){
                System.out.print(character + " ");
                character--;
            }
            System.out.println();
        }
    }
}
```

Output:

```
E
E D
E D C
E D C B
E D C B A
```

Character Pyramid Pattern

Description: Characters increase then decrease in pyramid form

Code:

```
class Main {
    public static void main(String[] args) {
        int num = 4;
        for (int row=1; row<= num; row++){
            for (int col = 1 ; col <= num-row; col++){
                System.out.print(" ");
            }
            char character = 65;
            for (int col = 1 ; col <= row; col++){
                System.out.print(character + " ");
                character++;
            }
        }
    }
}
```

```
    }  
    char characterI = (char)(row + 65 - 2);  
    for (int col = 1 ; col <= row-1; col++){  
        System.out.print(characterI + " ");  
        characterI--;  
    }  
    System.out.println();  
}  
}
```

Output:

```
    A  
  A B A  
A B C B A  
A B C D C B A
```